

Metabolism and Bioenergetics:

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mt-Nd2^a Suppresses Reactive Oxygen Species Production by Mitochondrial Complexes I and III*[S]

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Reactive oxygen species (ROS) play a critical role in the pathogenesis of human diseases. A cytosine to adenine transversion in the mitochondrially encoded NADH dehydrogenase subunit 2 (*mt-ND2*, human; *mt-Nd2*, mouse) gene results in resistance against type 1 diabetes and several additional ROS-associated conditions. Our previous studies have demonstrated that the adenine-containing allele (*mt-Nd2*^{a}) is also strongly associated with resistance against type 1 diabetes in mice. In this report we have confirmed that the cytosine-containing allele (*mt-Nd2*^{c}) results in elevated mitochondrial ROS production. Using inhibitors of the electron transport chain, we show that when in combination with nuclear genes from the alloxan-resistant (ALR) strain, *mt-Nd2*^{c} increases ROS from complex III. Furthermore, by using alamethicin-permeabilized mitochondria, we measured a significant increase in electron transport chain-dependent ROS production from all *mt-Nd2*^{c}-encoding strains including ALR.mt^{NOD}, non-obese diabetic (NOD), and C57BL/6 (B6). Studies employing alamethicin and inhibitors were able to again localize the heightened ROS production in ALR.mt^{NOD} to complex III and identified complex I as the site of elevated ROS production from NOD and B6 mitochondria. Using submitochondrial particles, we confirmed that in the context of the NOD or B6 nuclear genomes, *mt-Nd2*^{c} elevates complex I-specific ROS production. In all assays mitochondria from *mt-Nd2*^{a}-encoding strains exhibited low ROS production. Our data suggest that lowering overall mitochondrial ROS production is a key mechanism of disease protection provided by *mt-Nd2*^{a}.

Reactive oxygen species (ROS)[2] play an important role in the pathologies of many diseases such as atherosclerosis (1–5), myocardial infarction (6, 7), and hypertension (8, 9) due to the ability of ROS to damage lipids, protein, and DNA. ROS production has also been associated with the process of aging (10–12). Indeed, it has been reported that overexpression of the H_2O_2-detoxifying enzyme catalase resulted in increased life span in mice (13). It is believed that these aging-associated ROS are responsible for impaired cognitive function and neurodegeneration (14, 15) and play a role in the etiologies of Alzheimer (16) and Parkinson (17) diseases. Importantly, increased ROS production and decreased levels of reduced glutathione (GSH) as well as reduced transcripts of catalase, superoxide dismutase, and thioredoxin (18–20) have also been shown to play a key role in the destruction of β cells during the pathogenesis of type 1 diabetes (T1D).

Mitochondria represent the primary source of intracellular-generated ROS (21). During the process of mitochondrial oxidative phosphorylation, between 0.4 and 4% of oxygen is univalently reduced to superoxide (O_2^-) (22–24). Manganese superoxide dismutase converts O_2^- to H_2O_2, which is converted to the more deleterious hydroxyl radical via the Fenton reaction (25). NADH: ubiquinone oxidoreductase (complex I) and cytochrome *c* reductase (complex III) of the electron transport chain have been shown to be the major sites of mitochondrial ROS production (26, 27). Mutations in the mitochondria DNA (mtDNA) have been associated with a variety of human pathologies, including aging (28, 29) and cancer (30, 31). However, a link between specific mtDNA polymorphisms and mitochondria ROS production has yet to be reported in humans.

A single nucleotide polymorphism (C5173A), resulting in a leucine to methionine amino acid substitution in the mitochondrially encoded NADH dehydrogenase subunit 2 (*mt-ND2*) gene, has been associated with resistance against T1D in an autoantibody-positive Japanese population (32). In addition to T1D, this allele (*mt-Nd2*^{a}) has been reported to provide resistance against other ROS-mediated pathologies: atherosclerosis (33), myocardial infarction (34, 35), and high blood pressure (36)), and correlates with increased longevity (37) and reduced serum lipid levels (38). We hypothesize that decreased mitochondrial ROS production mediated by *mt-Nd2*^{a} plays a key role in this allele's protective effects.

Recently, we have shown that an orthologous single nucleotide polymorphism (C4738A), also resulting in a leucine to methionine substitution, encoded by the alloxan-resistant (ALR) mouse strain, provides resistance against the development of spontaneous T1D compared with the T1D prone non-obese diabetic (NOD) mouse strain (39). In these mice ALR

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[S] The on-line version of this article (available at http://www.jbc.org) contains supplemental Figs. 1–4.

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[2] The abbreviations used are: ROS, reactive oxygen species; *mt-ND2*, human NADH dehydrogenase subunit 2 gene; *mt-Nd2*, mouse NADH dehydrogenase subunit 2 gene; *mt-Nd2*^{a}, adenine encoding NADH dehydrogenase gene; *mt-Nd2*^{c}, cytosine encoding NADH dehydrogenase gene; complex I, NADH dehydrogenase; complex III, cytochrome *c* reductase; ALR, alloxan-resistant mouse strain; NOD, non-obese diabetic mouse strain; NOD.mt^{ALR}, conplastic mouse strain with NOD nuclear DNA and ALR mitochondrial DNA; ALR.mt^{NOD}, conplastic mouse strain with ALR nuclear DNA and NOD mitochondrial DNA; complex II, succinate dehydrogenase; complex IV, cytochrome *c* oxidase; T1D, type 1 diabetes.

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encodes the adenine-containing allele (*mt-Nd2*^{a}), and NOD encodes the cytosine-containing allele (*mt-Nd2*^{c}). In reciprocal backcrosses between NOD and ALR, the presence of *mt-Nd2*^{a} resulted in a 4-fold lower incidence of T1D (39). Along with *mt-Nd2*^{a}, loci on chromosomes 3, 8, and 17 have been mapped to provide resistance against T1D in ALR (40). Conplastic mouse strains were generated to better study the effect of allelic variants of *mt-Nd2*, one with NOD nuclear DNA and ALR mtDNA (NOD.mt^{ALR}) and one with ALR nuclear DNA and NOD mtDNA (ALR.mt^{NOD}) (41).

We have demonstrated that this single nucleotide polymorphism in *mt-Nd2* has no effect on mitochondrial respiration, membrane potential, or electron transport chain enzymatic activities (41). Although it was hypothesized that this single nucleotide polymorphism may itself be involved in the dissipation of ROS (32), our results did not support this postulate as exogenously added free radicals affected *mt-Nd2*^{a} and *mt-Nd2*^{c} encoding mitochondria equally (41). However, *mt-Nd2*^{c} in combination with ALR nuclear DNA resulted in increased reactive oxygen species production from ALR.mt^{NOD} mitochondria (41). This is in agreement with a previous report in which cybrid cells were created containing identical nuclear DNA but various common mitochondrial haplotypes (42). This paper from Moreno-Loshuertos *et al.* (42) demonstrated that respiration among these different cybrid cell lines did not vary; however, there were compensatory alterations in whole cell ROS production.

In this study we set out to further characterize the suppression of ROS production due to *mt-Nd2*^{a} and the elevated ROS production by *mt-Nd2*^{c}. Using specific inhibitors of the electron transport chain, we demonstrate that increased ROS production from ALR.mt^{NOD} mitochondria is generated from complex III. Additionally, we show that when *mt-Nd2*^{c} is in combination with the NOD and B6 genomes, mitochondria generated elevated ROS from complex I, which produces ROS primarily into the mitochondrial matrix. We conclude that *mt-Nd2*^{a} results in an attenuated mitochondrial ROS generation from both mitochondrial complexes I and III, independently of nuclear genome.

EXPERIMENTAL PROCEDURES

Mice—Selection of the included mouse strains was based on mitochondrial genotype (Table 1). All breeding and care of ALR/LtJ, NOD/ShiLtJ, C57BL/6J (B6), and NOD.129S7(B6)-*Rag1*$^{tm1Mom/J}$ (NOD.*Rag1*$^{-/-}$) as well as the reciprocal conplastic mouse strains, NOD/ShiLtJ-mt$^{ALR/LtJ}$/Mx (NOD.mt^{ALR}) and ALR/LtJ-mt$^{NOD/ShiLtDvs}$/Mx (ALR.mt^{NOD}), generated as described (41), were conducted in a specific pathogen-free vivarium. Mice were allowed free access to food (autoclaved diet NIH-31, 6% fat, PMI, St. Louis, MO) and acidified drinking water at the animal research facilities of both the Rangos Research Center, Pittsburgh, PA, and the University of Florida. Children's Hospital of Pittsburgh and University of Florida approvals were obtained for all procedures involving animals, and each procedure was in compliance with the "Principles of Laboratory Animal Care" and the current laws of the United States of America.

Mitochondrial Isolation—Mice were euthanized, and livers were promptly removed. Homogenization was carried out in 15-ml Dounce homogenizers (Fisher) in isolation buffer I (225 mM mannitol, 75 mM sucrose, 10 mM HEPES potassium salt, 0.10% fatty acid free bovine serum albumin, and 1 mM EDTA, at pH 7.4). The homogenate was transferred into clean centrifuge tubes and centrifuged at 1300 × *g* for 10 min. The supernatants were decanted into new tubes and centrifuged at 10,000 × *g* for 10 min. The resulting supernatants were discarded, and the pellets were resuspended in isolation buffer II (225 mM mannitol, 75 mM sucrose, 10 mM HEPES potassium salt, 100 μM EDTA, at pH 7.4) and centrifuged at 10,000 × *g* for 10 min. The resulting supernatant was discarded, and the pellet was resuspended in roughly 100 μl of isolation buffer II. Protein concentration was subsequently determined using the BCA protein assay (Pierce).

Isolation of Submitochondrial Particles—Submitochondrial particles were isolated as described previously (27). The pellet of submitochondrial particles was resuspended in ~100 μl of isolation buffer II. Protein concentration was determined using the BCA protein assay.

Measurement of Complex I Activity and Complex I to Complex III Electron Shuttling by Alamethicin-permeated Mitochondria or SMPs—Isolated mitochondria incubated with alamethicin (30 μg/ml) or submitochondrial particles were assayed for activity of complex I was determined by monitoring the oxidation of NADH at 340 nm as described previously (41). The electron shuttling activity between complexes I and III was determined in either isolated mitochondria incubated with alamethicin (30 μg/ml) or submitochondrial particles by monitoring the reduction of ferricytochrome *c* at 550 nm. The assay medium contained potassium phosphate (25 mM, pH 7.2 at 20 °C), 5 mM $MgCl_2$, 2.5 mg/ml bovine serum albumin (fraction V), and 2 mM KCN. KCN was included in the assay media to prevent the reoxidation of the production, ferrocytochrome *c*, by cytochrome *c* oxidase. Nonenzymatic activity was recorded for 1 min after the addition of 15 μM ferricytochrome *c*, 0.13 mM NADH, and 30 μg/ml alamethicin. The complex I-III electron shuttling activity was initiated by the addition of freshly isolated, coupled mitochondria (200 μg/ml), and the rate of reduction of ferricytochrome *c* to ferrocytochrome *c* was recorded for 3 min. In replicate wells 2 μg/ml antimycin A was added, and the complex I-III-specific activity was calculated by subtracting the antimycin A-insensitive activity from the total activity.

Mitochondrial ROS Production—Mitochondrial ROS production was assessed as previously described using a Shimadzu RF-5301 spectrofluorimeter (Kyoto, Japan) (44). All assays were performed with constant stirring at 37 °C using 300 μg of mitochondrial protein in incubation media supplemented with either 5 mM L-glutamate and 5 mM L-malate or 5 mM succinate. ROS production was measured by continuously recording the real-time oxidation of 2 μM Amplex Red dye (Molecular Probes, Eugene, OR) to the fluorescent molecule resorufin, catalyzed by 1 unit/ml horseradish peroxidase. Fluorescence was measured with an excitation wavelength of 560 nm (slit 1.5 nm) and an emission wavelength of 590 nm (slit 3 nm). Rates of ROS production were measured for at least 2 min for each treatment. All rates were linear over the time interval measured. Slopes were converted into units of pmol H_2O_2/min/mg as described (45).

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ROS Production by Intact Mitochondria in the Presence of Electron Transport Chain Inhibitors—ROS production was assessed in the presence of various inhibitors of the electron transport chain to determine the specific site of increased ROS. Using intact mitochondria (300 μg), basal ROS production was first assessed for 2 min in the presence of either complex I substrates (glutamate, malate, and malonate; 5 mM each) or a succinate dehydrogenase (complex II) substrate (succinate, 5 mM). ROS production supported by complex I substrates was then assessed in the presence of rotenone (10 μM), myxothiazol (10 μM), antimycin A (10 μM), or potassium cyanide (10 μM) for 2 min after the addition of each inhibitor. ROS production supported by complex II substrates was assessed in the presence of rotenone (10 μM), myxothiazol (10 μM), antimycin A (10 μM), potassium cyanide (10 μM), or rotenone and myxothiazol (10 μM, each) for 2 min after the addition of each inhibitor. All rates were linear over the time interval measured (supplemental Figs. 1 and 2).

ROS Production by Intact Mitochondria in the Presence of Alamethicin—For control traces, mitochondrial ROS production (200 μg of intact mitochondria) was assessed for 2 min with no added substrates followed by the addition of NADH (80 μM) for an additional 2 min. For experimental traces, mitochondrial ROS production was first assessed in the presence of alamethicin (30 μg/ml) with no added substrates for 2 min. NADH (80 μM) was added, and ROS production was assessed for 2 min. ROS production was measured for an additional 2 min with the addition of rotenone (10 μM), rotenone and *p*-chloromercuribenzoate (CMB) (10 μM, each), or myxothiazol (10 μM). All reactions were conducted in the presence of 40 units/ml superoxide dismutase due to the previously reported ability of reduced pyridine nucleotides to oxidize Amplex Red in a superoxide-dependent fashion (45). All rates were linear over the time interval measured (supplemental Fig. 3).

ROS Production by Submitochondrial Particles—After the addition of submitochondrial particles (75 μg), ROS production was assessed for 2 min without added substrates and for 2 min with the addition of either NADH (80 μM) or succinate and rotenone (5 mM and 10 μM, respectively). ROS production was measured for an additional 2 min after the addition of CMB (10 μM). All reactions contained 40 units/ml superoxide dismutase. ROS production was linear over the 2-min measurement periods (supplemental Fig. 4).

RESULTS

Site of Elevated ROS Production in Intact ALR.mt^{NOD} Mitochondria—The mitochondrial genotypes for all mouse strains included in this work are in Table 1 (39, 41). Previously, we have reported that isolated and intact ALR.mt^{NOD} mitochondria produce elevated ROS in comparison to ALR, NOD, NOD.mt^{ALR}, and B6 (41). NOD.*Rag1*$^{-/-}$ intact mitochondria produce ROS equivalently to ALR, NOD, NOD.mt^{ALR}, and B6 (data not shown). In the current study we used specific inhibitors of the electron transport chain to determine the site of increased ROS in intact ALR.mt^{NOD} mitochondria. In Fig. 1*A* and Fig. 2*A* (supplemental Figs. 1 and 2), we confirm that ALR.mt^{NOD} mitochondria produce ~30% more ROS supported by either complex I or complex II substrates than ALR,

TABLE 1

Comparison of the *mt-Nd2* and *mt-Tr* alleles encoded by experimental mouse strains

The genotype of each mouse strain is listed for *mt-Nd2* at base pair 4738 and for *mt-Tr* at base pair 9827 in the mtDNA. C4738 encodes leucine, whereas A4738 encodes methionine.

Strain	*mt-Nd2*	*mt-Tr*
ALR/LtJ (ALR)	A	9A
NOD/ShiLtJ-mt$^{ALR/LtJ}$/Mx (NOD.mt^{ALR})	A	9A
ALR/LtJ-mt$^{NOD/ShiLtDvs}$/Mx (ALR.mt^{NOD})	C	9A
NOD/ShiLtJ (NOD)	C	10A
NOD.129S7(B6)-*Rag1*$^{tm1Mom/J}$ (NOD.*Rag1*$^{-/-}$)	C	9A
C57BL/6J (B6)	C	8A

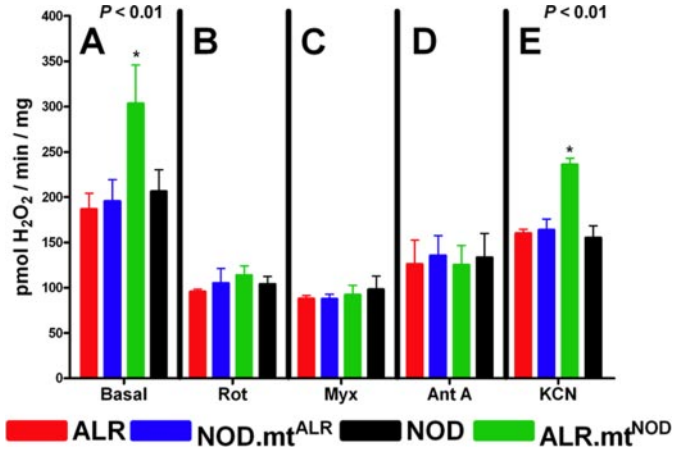


FIGURE 1. **Complex III redox centers are responsible for elevated ROS production from intact ALR.mt^{NOD} mitochondria respiring on complex I substrates.** All reactions contained the complex I substrates glutamate (5 mM) and malate (5 mM) and the complex II inhibitor malonate (5 mM). Reactions were initiated by the addition of 300 μg of mitochondrial protein. After 3 min of continuously recording the rate of basal ROS production (*A*), rotenone (*Rot*, 10 μM) (*B*), myxothiazol (*Myx*, 10 μM) (*C*), antimycin A (*Ant A*, 10 μM) (*D*), or KCN (10 μM) (*E*) were added to the reaction. The rate of ROS production was measured for 2 min after the addition of each inhibitor. $n \geq 4$ for mitochondria from all strains.

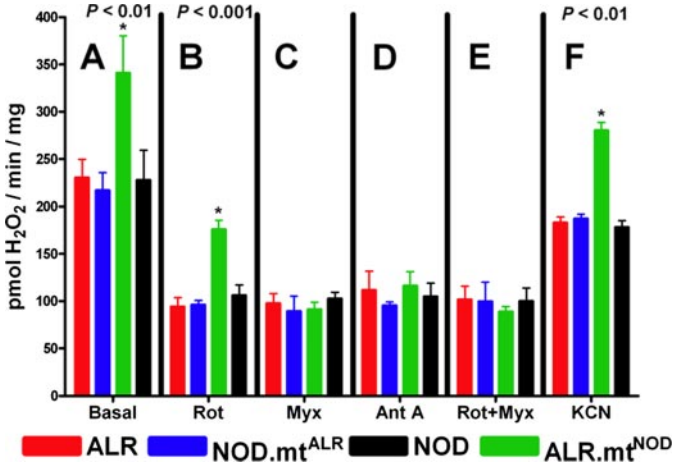


FIGURE 2. **Complex III redox centers are responsible for elevated ROS production from intact ALR.mt^{NOD} mitochondria respiring on complex II substrates.** All reactions contained the complex II substrate succinate (5 mM). Reactions were initiated by the addition of 300 μg of mitochondrial protein. After 2 min of continuously recording the rate of basal ROS production (*A*), rotenone (*Rot*, 10 μM) (*B*), myxothiazol (*Myx*, 10 μM) (*C*), antimycin A (*Act A*, 10 μM) (*D*), rotenone and myxothiazol (10 μM each) (*E*), or KCN (10 μM) (*F*) were added to each reaction. The rate of ROS production was measured for 2 min after the addition of each inhibitor. $n \geq 4$ for mitochondria from all strains.

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NOD, and NOD.mt^{ALR} mitochondria ($p < 0.01$). Rotenone (10 μM), which blocks electron flow in complex I near the ubiquinone binding site (46), was added to mitochondria respiring on complex I substrates. Rotenone significantly reduced ROS production by intact mitochondria from all strains of mice and abrogated the elevated ROS production in ALR.mt^{NOD} mitochondria (Fig. 1*B*, supplemental Fig. 1*A*). Furthermore, the complex III inhibitors myxothiazol and antimycin A (10 μM each) both significantly reduced ROS production and eliminated the elevation in ROS production from intact ALR.mt^{NOD} mitochondria (Figs. 1, *C* and *D*, supplemental Figs. 1, *B* and *C*). The complex IV inhibitor KCN (10 μM) decreased ROS production from all strains; however, ALR.mt^{NOD} maintained its elevated ROS production ($p < 0.01$) (Fig. 1*E*, supplemental Fig. 1*D*). While producing ROS supported by complex II substrates, rotenone reduced ROS production in mitochondria from all mouse strains, but ALR.mt^{NOD} mitochondria retained its increased ROS production ($p < 0.001$) (Fig. 2*B*, supplemental Fig. 2*A*). This increase was abrogated by the addition of myxothiazol after rotenone (Fig. 2*E*, supplemental Fig. 2*A*). Similarly to complex I substrates, the addition of myxothiazol or antimycin A to mitochondria respiring on complex II substrates eliminated the elevated ROS production in ALR.mt^{NOD} mitochondria (Figs. 2, *C* and *D*, supplemental Fig. 2, *B* and *C*). KCN did not eliminate this difference ($p < 0.01$) (Fig. 2*F*, supplemental Fig. 2D).

Detection of Total ROS Produced by Mitochondria—The system to detect mitochondrial ROS production we employed above senses ROS produced into the intermembrane space. Both complexes I and III produce ROS into the mitochondrial matrix (27, 47, 48). To assess the production of ROS into the matrix, we used alamethicin to form pores in the inner membrane allowing for the diffusion of superoxide from the matrix into the intermembrane space. Complex I and III inhibitors were used to determine the sites of elevated ROS production. This system allows complex I ROS production to be directly supported by the addition of NADH. ROS production was measured without NADH or alamethicin (Fig. 3*A*), with NADH but without alamethicin (Fig. 3*B*), and without NADH but with alamethicin (Fig. 3*C*) as controls (supplemental Fig. 3*A*). As seen in Fig. 2*D* (supplemental Fig. 3*B*), ROS production in the presence of NADH and alamethicin is significantly increased in ALR.mt^{NOD}, NOD, NOD.*Rag1*$^{-/-}$, and B6 mitochondria ($p < 0.01$). As expected, the addition of rotenone eliminated the increased ROS production in ALR.mt^{NOD} mitochondria; however ROS production from NOD, NOD.*Rag1*$^{-/-}$, and B6 mitochondria remained significantly elevated ($p < 0.05$) (Fig. 3*E*, supplemental Fig. 3*B*). Adding CMB, an inhibitor of complex I iron-sulfur clusters (49), eliminated the elevated ROS production from NOD, NOD.*Rag1*$^{-/-}$, and B6 mitochondria (Fig. 3*F*, supplemental Fig. 3*B*). Consistently, the addition of myxothiazol eliminated the increased ROS production from ALR.mt^{NOD} mitochondria but not from *mt-Nd2*^{c} encoding mitochondria ($p < 0.05$) (Fig. 3*G*, supplemental Fig. 3*C*).

ROS Production in Submitochondrial Particles—To simplify our ROS-producing system, we prepared submitochondrial particles containing only the membrane-bound enzymes of the mitochondrial electron transport chain (27). Importantly, we

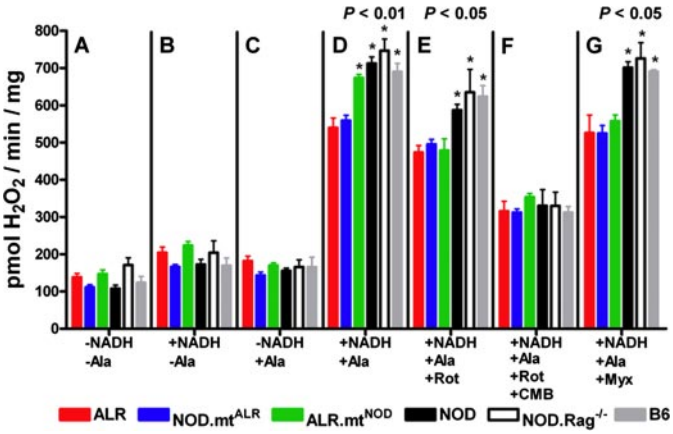


FIGURE 3. **Complex I produces elevated ROS released into the mitochondrial matrix from NOD and B6 mitochondria.** Reactions were initiated by the addition of 300 μg of mitochondrial protein. All reactions contained superoxide dismutase (40 units/ml) to eliminate the superoxide-dependent pyridine nucleotide-mediated oxidation of Amplex Red. For controls, the rate of ROS production was assessed without NADH or alamethicin (*Ala*, *A*), with NADH (80 μM) and without alamethicin (*B*), and without NADH and with alamethicin (30 μg/ml) (*C*). To directly measure complex I ROS production, NADH (80 μM) was added to mitochondria permeabilized by alamethicin (30 μg/ml) (*D*). After continuously recording NADH stimulated ROS production for 2 min, rotenone (*Rot*, 10 μM) (*E*), myxothiazol (*Myx*, 10 μM) (*G*), or rotenone followed by CMB (10 μM, each) (*F*) were added to the reaction for an additional 2 min. $n \geq 6$ for ALR, NOD.mt^{ALR}, ALR.mt^{NOD}, and NOD mitochondria; $n = 3$ for NOD.*Rag1*$^{-/-}$ and B6 mitochondria.

prepared non-phosphorylating submitochondrial particles (50) that have significantly decreased electron flow between complex I and III. The reduction of cytochrome *c* by either alamethicin-permeabilized B6 mitochondria or B6 submitochondrial particles ($n = 4$) supplemented with NADH was compared. Alamethicin-permeabilized B6 mitochondria exhibited a significantly higher cytochrome *c* reduction (43.30 $\pm$ 3.57 nmol/min/mg) compared with B6 submitochondrial particles (11.21 $\pm$ 2.28 nmol/min/mg). However, complex I activity alone in these submitochondrial particles (37.37 $\pm$ 2.48 nmol of NADH oxidized/min/mg) was similar to our previously reported B6 complex I activity (34.70 $\pm$ 6.04 nmol NADH oxidized/min/mg). Therefore, although these non-phosphorylating submitochondrial particle preparations maintain high complex I activity, electron shuttling between complexes I and III is decreased by 75%. As shown in Fig. 4*A* as well as supplemental Fig. 4*A*, ROS production by submitochondrial particles supported by NADH (80 μM) was significantly higher in NOD, B6, and NOD.*Rag1*$^{-/-}$ submitochondrial particles compared with ALR, ALR.mt^{NOD}, and NOD.mt^{ALR} submitochondrial particles ($p < 0.01$). ROS production supported by complex II substrates showed no difference among submitochondrial particles from the different strains (Fig. 4*B*, supplemental Fig. 4*B*). The addition of CMB to submitochondrial particles oxidizing NADH ablated the elevated ROS production from NOD, B6, and NOD.*Rag1*$^{-/-}$ mitochondria (Fig. 4*C*, supplemental Fig. 4*A*). CMB had no significant effect on ROS produced by submitochondrial particles oxidizing complex II substrates (Fig. 4*D*, supplemental Fig. 4*B*).

DISCUSSION

mt-Nd2^{a} has been shown to provide resistance against T1D in humans and mice as well as against several other ROS-mediated conditions (32–39). This allele is also associated with

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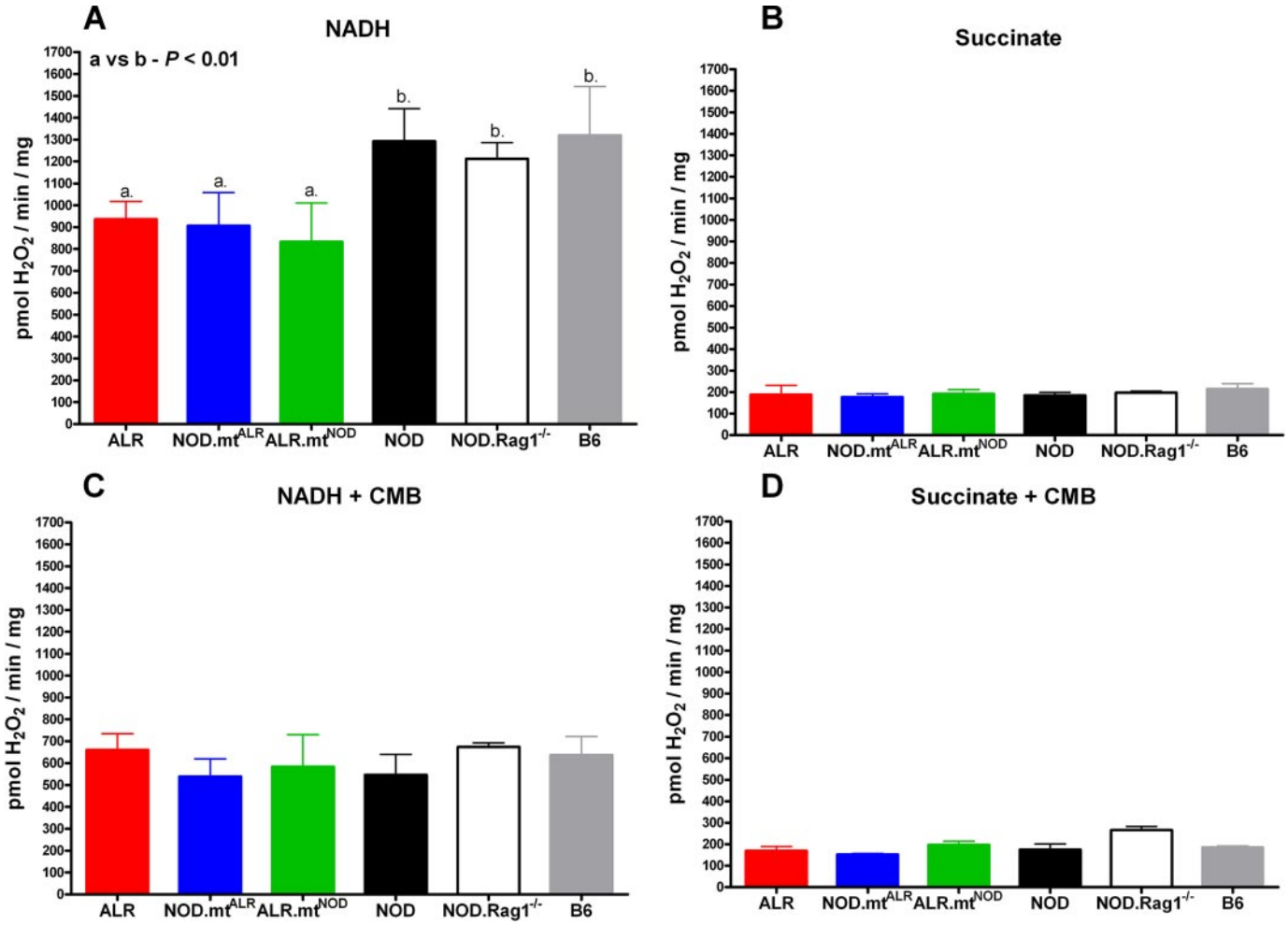


FIGURE 4. **Submitochondrial particle ROS production confirms that *mt-Nd2^c* results in elevated ROS production from complex I.** Reactions were initiated by the addition of 75 μg of submitochondrial particle protein. *A*, the rate of ROS production was assessed supported by NADH (80 μg) while inhibiting complex II with malonate (5 mM) for 3 min followed by the addition of CMB (10 μM) for 2 min (*C*). All reactions using NADH also contained superoxide dismutase (40 units/ml). *B*, ROS production was assessed supported by succinate (5 mM) while inhibiting complex I with rotenone (10 μM) for 3 min followed by the addition of CMB (10 μM) for 2 min (*D*). $n \geq 8$ for ALR, NOD.mt^{ALR}, ALR.mt^{NOD}, NOD, and B6 mitochondria; $n = 3$ for NOD.*Rag1*$^{-/-}$ mitochondria.

resistance to alloxan-induced, free radical-mediated diabetes (39). We have previously reported that the common allele, *mt-Nd2^c*, in combination with ALR nuclear DNA results in elevated mitochondrial ROS production (41). In this report we have further characterized the effect of *mt-Nd2^a* on ROS production from isolated mitochondria and determined the sites of elevated ROS production in the context of *mt-Nd2^c*.

Previous studies have detailed the sites of ROS production from isolated mitochondria. Liver mitochondria produce ROS and respond to electron transport chain inhibitors differently when compared with the more extensively studied brain, heart, and skeletal mitochondria (51–55). In brain, heart, and skeletal muscle mitochondria, rotenone accelerates ROS production while respiring on complex I substrates (51), and myxothiazol and antimycin A increase ROS production while respiring on complex I or complex II substrates (52–54). Liver mitochondria exhibit a markedly different phenotype (55). In these mitochondria, respiring on either complex I or complex II substrates, rotenone, myxothiazol, and antimycin A all inhibit ROS production (Figs. 1 and 2). Furthermore, the site of ROS production from complex I in brain and heart mitochondria has been localized to the flavin mononucleotides and has been shown to be inhibited by diphenyleneiodonium chloride (48, 56). In liver mitochondria, we observe no inhibition of ROS production with diphenyleneiodonium chloride (data not shown) but observe drastic inhibition with CMB (Figs. 3 and 4), indicating that the site of ROS production in liver mitochondria may be the iron sulfur clusters.

To determine the site of the previously reported increased ROS production from ALR.mt^{NOD} mitochondria (41), we assessed ROS production in the presence of various inhibitors of the electron transport chain. Rotenone was shown to inhibit elevated ROS production from ALR.mt^{NOD} mitochondria supported by complex I substrates (Fig. 1*B*). Rotenone binds within complex I near the ubiquinone binding site (46, 57, 58), allowing the ROS-producing redox centers of complex I to remain active (48). While supported by complex II substrates alone, electrons are donated directly from complex II to complex III, and complex I redox centers are only utilized due to reverse electron flow from complex III to complex I (26), which can be blocked by the addition of rotenone. This is evidenced by the reduction in overall succinate-supported ROS production in all

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of the strains after the addition of rotenone (Fig. 2, *A versus B*). The elevated ROS signature from ALR.mt^{NOD} mitochondria was still maintained in the presence of rotenone supported by complex II substrates (Fig. 2*B*). These results indicate that the increased ROS production from ALR.mt^{NOD} mitochondria is generated downstream of the complex I redox centers. ROS production was also assessed with the addition of the complex III inhibitors myxothiazol and antimycin A (Figs. 1, *C* and *D*, and 2, *C* and *D*). Antimycin A binding overlaps with the Q_i (quinine reduction) site of the bc_1 complex, whereas myxothiazol binds the Q_o (quinol oxidation) site of the bc_1 complex (59). Thus, these inhibitors block electron flow on opposite sides of the inner membrane, with antimycin A blocking electron flow from heme b_H to ubiquinone and myxothiazol blocking electron flow from heme b_L to the iron-sulfur cluster (59). When supported by either complex I or II substrates, both myxothiazol and antimycin A inhibited the elevated ROS production from ALR.mt^{NOD} mitochondria (Figs. 1, *C* and *D*, and 2, *C* and *D*). Furthermore, increased ROS production from ALR.mt^{NOD} mitochondria supported by either complex I or complex II substrates was maintained after the addition of the cytochrome *c* oxidase (complex IV) inhibitor KCN (Figs. 1*E* and 2*F*). This demonstrates that the site of elevated ROS production is likely the Q cycle of complex III.

Considering that *mt-Nd2* encodes a complex I protein, it is surprising that complex III ROS production is affected by its allelic variants. We propose that *mt-Nd2*^{c} acts in concert with one or more ALR-derived nuclear loci to alter the production of ROS from complex III. Several reports have shown that respiratory supercomplexes form between complexes I and III as well as complexes I, III, and IV (60–62). These supercomplexes may better stabilize the electron transport chain and allow for more efficient electron transfer between the complexes (60–62). Combined with components of complex III encoded by ALR nuclear DNA, *mt-Nd2*^{c} may act to alter either supercomplex formation between complexes I and III or the structure of the supercomplex. Subsequent complex instability and reduced electron transfer efficiency may facilitate the donation of electrons to molecular oxygen, resulting in elevated ROS production from complex III.

Although we have shown that allelic variants of *mt-Nd2* can affect complex III ROS production, our experimental system using intact mitochondria did not allow us to directly assess the effect of *mt-Nd2* on complex I ROS production. ROS are generated by complex III at both Q_o, into the intermembrane space, and by Q_o, into the matrix (27, 47). Measurement of ROS production from complex III or complex I into the matrix can only be accurately performed if the mitochondrial inner membrane is made permeable. The oxidation of Amplex Red to resorufin used for detection of ROS in this study depends on the conversion of superoxide, produced by complex I or complex III, to hydrogen peroxide by superoxide dismutase (27, 51). Hydrogen peroxide can freely diffuse across the mitochondrial inner membrane, whereas the negatively charged superoxide anion and Amplex Red cannot (43). To more directly assess ROS production from complex I, alamethicin was used to introduce channels in the inner membrane and allow the diffusion of NADH and superoxide between the matrix and inner-membrane space. Exogenous Cu,Zn-superoxide dismutase was added to each reaction to eliminate the superoxide-mediated, pyridine nucleotide-dependent oxidation of Amplex Red. Adding excess superoxide dismutase also assured that all superoxide generated by the electron transport chain was converted into hydrogen peroxide.

Stimulating complex I with the complex I substrate NADH after treatment with alamethicin revealed that not only ALR.mt^{NOD}, but also NOD, NOD.*Rag1*$^{-/-}$, and B6 mitochondria produced significantly more ROS than ALR and NOD.mt^{ALR} (Fig. 3*D*). In line with the findings from intact mitochondria, the increased ROS production from ALR.mt^{NOD} mitochondria was eliminated by the addition of rotenone or myxothiazol; however, neither rotenone nor myxothiazol inhibited the increased ROS production from NOD, NOD.*Rag1*$^{-/-}$, or B6 mitochondria (Figs. 3, *E* and *G*). The complex I iron-sulfur cluster inhibitor, CMB, eliminated elevated ROS production from NOD, NOD.*Rag1*$^{-/-}$, and B6, demonstrating that this increased ROS is derived from complex I redox centers (Fig. 3*F*). These findings support the conclusion that *mt-Nd2*^{c} increases complex I ROS, which is undetectable in intact mitochondria.

To confirm this phenotype, we prepared non-phosphorylating submitochondrial particles, eliminating soluble enzymes within the mitochondrial matrix or intermembrane space that may affect the measured rates of ROS production. Consistent with alamethicin-permeabilized mitochondria, NOD, NOD.*Rag1*$^{-/-}$, and B6 submitochondrial particles supported by NADH produced significantly more ROS than ALR, NOD.mt^{ALR}, and ALR.mt^{NOD} (Fig. 4*A*). ALR.mt^{NOD} submitchondrial particles do not produce increased ROS supported by NADH because electron flow from complex I to complex III is greatly reduced in our submitochondrial particles ("ROS Production in Submitochondrial Particles" under "Results"). Importantly, elevated ROS production by *mt-Nd2*^{c} encoding NOD and NOD.*Rag1*$^{-/-}$ mitochondria compared with NOD.mt^{ALR} indicates that ND2^{c} and not components encoded by the NOD nuclear DNA is responsible for elevated complex I ROS production. As with alamethicin-permeabilized mitochondria, CMB inhibited the increased ROS production in NOD, NOD.*Rag1*$^{-/-}$, and B6 submitochondrial particles (Fig. 4*C*). Succinate supported low levels of ROS production from submitochondrial particles, and no difference in ROS production was observed (Fig. 4*B*). ALR.mt^{NOD} did not produce elevated ROS likely because of decreased electron flow to complex III. CMB did not affect succinate-supported ROS production, supporting its complex I specificity (Fig. 4*D*). These results indicate that *mt-Nd2*^{c} does in fact result in increased ROS production from mitochondrial complex I when combined with either the NOD or B6 nuclear genome. In both alamethicin-permeabilized mitochondria and submitochondrial particles, ALR.mt^{NOD} did not produce increased ROS from complex I despite encoding *mt-Nd2*^{c}. These results clearly demonstrate the importance of interactions between alleles encoded by the nuclear genome and mtDNA in the modulation of mitochondrial ROS production.

Recently it has been reported that alleles of *mt-Tr*, the mitochondrial transfer RNA for arginine, have a role in cellular ROS

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TABLE 2

ROS production phenotypes exhibited by experimental mouse strains

The ROS production phenotype observing for each mouse strain from complex I (using alamethicin-permeabilized mitochondria or submitochondrial particles) and from complex III (using intact or alamethicin-permeabilized mitochondria) is listed above. High indicates a ROS production phenotype of roughly 30% greater than Low.

Strain	Complex I ROS	Complex III ROS
ALR	Low	Low
NOD.mt^{ALR}	Low	Low
ALR.mt^{NOD}	Low	High
NOD	High	Low
NOD.*Rag1*$^{-/-}$	High	Low
B6	High	Low

production by an undetermined mechanism (42). There are three common alleles of this locus (the variants differ by the number of adenines), either 8, 9, or 10 in the DHU loop of this tRNA. When cybrid cell lines that only differed by the *mt-Tr* allele were tested for ROS production, the cell lines with *mt-Tr$_{10}$* exhibited an ~35% increased ROS production compared with cybrid cells encoding *mt-Tr^9* or *mt-Tr^8* (42). To determine whether *mt-Tr* had an impact on mitochondrial ROS production in our systems, we tested mitochondria from NOD/LtJ mice (*mt-Tr^{10}*), NOD.*Rag1*$^{-/-}$ (*mt-Tr^9*), and B6 (*mt-Tr^8*) (39). All three strains encode *mt-Nd2^c* (Table 1). We did not measure any differences comparing these three mouse strains for ROS production from intact mitochondria (Figs. 1 and 2), alamethicin-permeated mitochondria (Fig. 3), or submitochondrial particles (Fig. 4).

In this report we demonstrate that *mt-Nd2* plays a critical role in the generation of mitochondrial ROS. Although *mt-Nd2^c* results in elevated ROS produced by either complex I or III, its action is dependent upon the nuclear genome with which it is combined. Conversely, *mt-Nd2^a* results in lowered ROS production regardless of nuclear DNA (Table 2). In the presence of *mt-Nd2^a*, neither complex I nor complex III ROS production was elevated in the context of either ALR or NOD nuclear DNA. We hypothesize that this lowered ROS production is a key mechanistic effect provided by *mt-Nd2^a* in resistance to various pathological conditions (1–13, 18). This hypothesis is supported by the finding that mice encoding *mt-Nd2^c* exhibit heightened susceptibility to alloxan-induced diabetes (39). By increasing intrinsic ROS production, *mt-Nd2^c* likely decreases the threshold of free radicals needed to induce cellular dysfunction. We suggest that in humans, *mt-ND2^a* will also lead to reduced mitochondrial ROS production, affording protection against ROS-associated pathologies.

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